Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/06/2008
PI Name:	McNeil, Paul L Ph.D.		
Project Title:	Cellular and Molecular Mechanisms Mediating Rapid Resealing of Gravity-Induced Plasma Membrane Disruptions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	pmcneil@mail.mcg.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(706) 721-3065
Organization Name:	Medical College of Georgia		
PI Address 1:	CA 2018		
PI Address 2:			
PI Web Page:			
City:	Athens	**State:**	GA
Zip Code:	30912	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC44G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The major goals of the research program remain unchanged: 1) to elucidate the organelle and molecular components used by skeletal muscle to repair tears in their surface plasma membrane; 2) to identify cell and molecular events leading to muscle hypertrophy		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Understanding of muscle disease.
Task Progress:	FINAL TASK BOOK REPORT The following highlights major progress/discoveries that were made possible by funding from NASA: 1) Development of a microscopic method for creating plasma membrane disruptions and monitoring repair of this injury quantitatively. This method is applicable to skeletal muscle, and so has provided insight into how muscle cells repair membrane tears, such as those that occur in astronauts when the reload under gravity muscles after a long space flight. McNeil PL, Miyake K, and Vogel SS. (2003) The endomembrane requirement for cell surface repair. PNAS, 100: 4592-4597. 2) This method was crucial to the discovery of a protein, dysferlin, required for skeletal muscle membrane repair. The paper reporting this result has already been cited over 130 times in the literature. Bansal D. Miyake K, Chen CC, Groh S, Williamson R, McNeil PL,(2003) and Campbell KP. Defective membrane repair in dysferlin-deficient muscular dystrophy. Nature, 423:168-172 3) Identification of two additional proteins required for repair, annexin and calpain. McNeil AK, Rescher U, Gerke V, and McNeil PL. (2006) Requirement for annexin A1 in plasma membrane repair. J Biol Chem., 281:35202-7. Mellgren RL, Zhang W, Miyake K, and McNeil PL. (2007). Calpain is Required for the Rapid, Calcium-dependent Repair of Wounded Plasma Membrane. J Biol. Chem.282:2567-75.
Bibliography Type:	Description: (Last Updated: 06/11/2008)
Articles in Peer-reviewed Journals	Mellgren RL, Zhang W, Miyake K, McNeil PL. "Calpain is required for the rapid, calcium-dependent repair of wounded plasma membrane." J Biol Chem. 2007 Jan 26;282(4):2567-75. Epub 2006 Nov 22. PMID: 17121849 , Jan-2007
Articles in Peer-reviewed Journals	McNeil AK, Rescher U, Gerke V, McNeil PL. "Requirement for annexin A1 in plasma membrane repair." J Biol Chem. 2006 Nov 17;281(46):35202-7. Epub 2006 Sep 19. PMID: 16984915 , Nov-2006